CSE220 Digital Audio Equalizer -- DSP Analysis Report

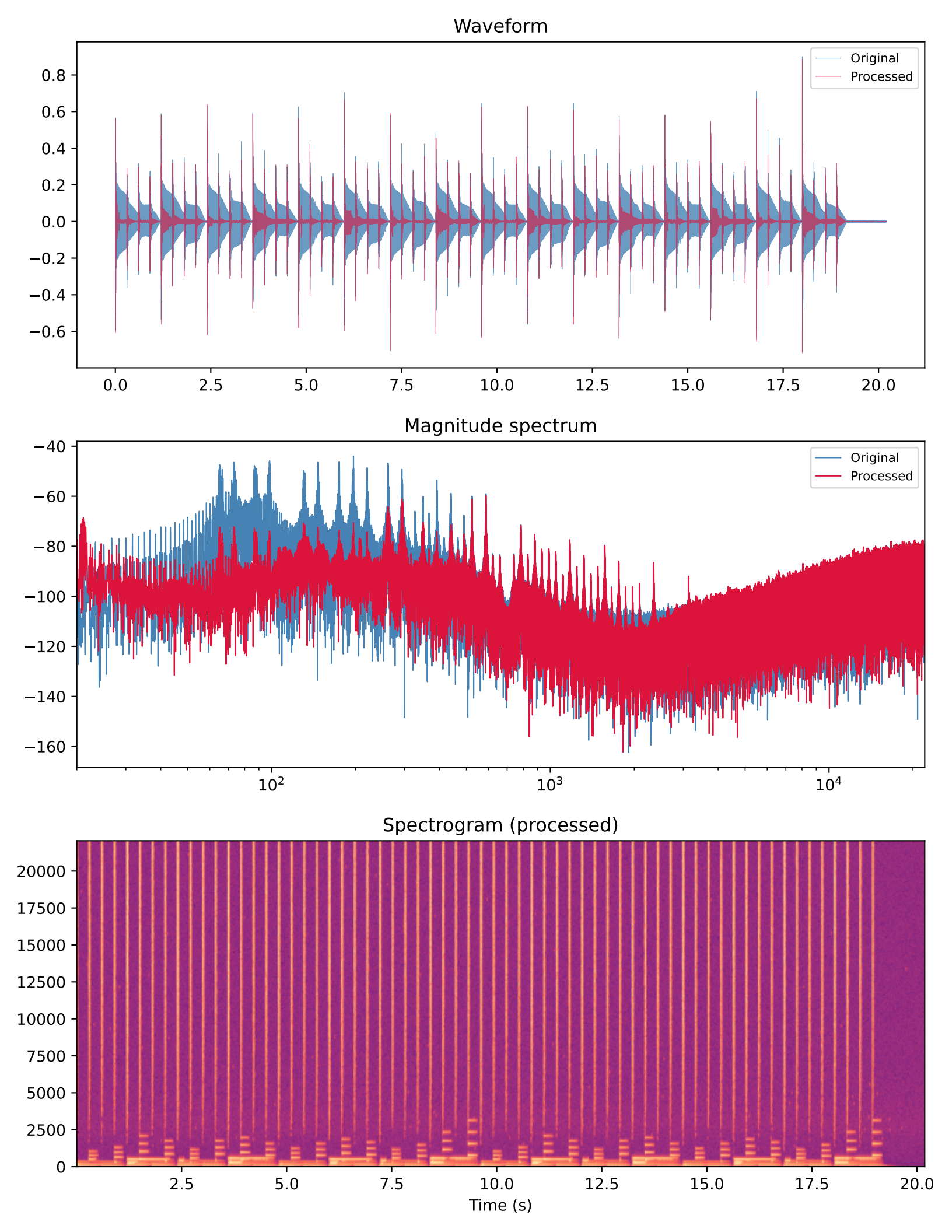
```
Mode: parametric
EQ settings: smoke test

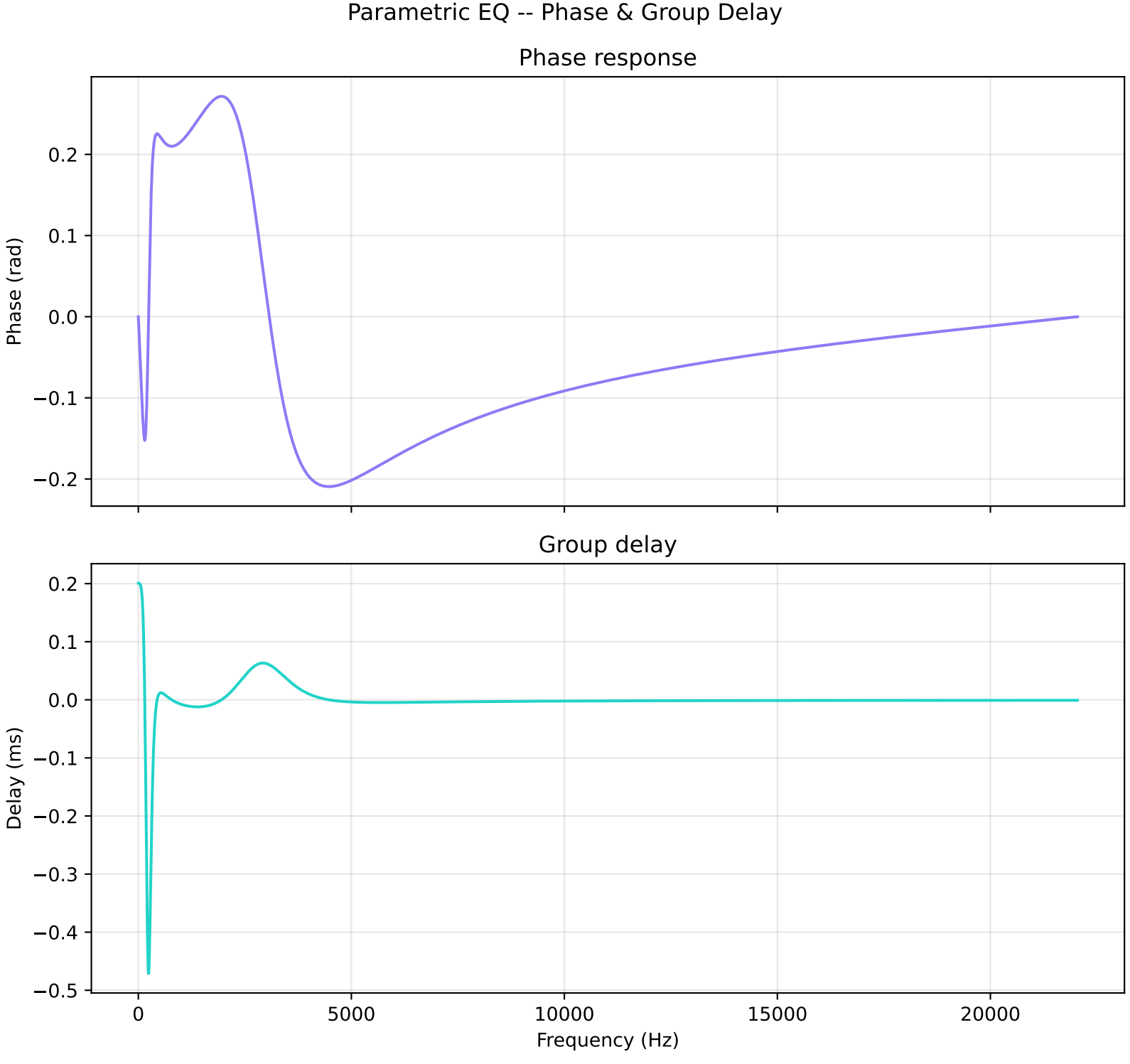
Sample rate: 44100 Hz
Duration: 20.20 s

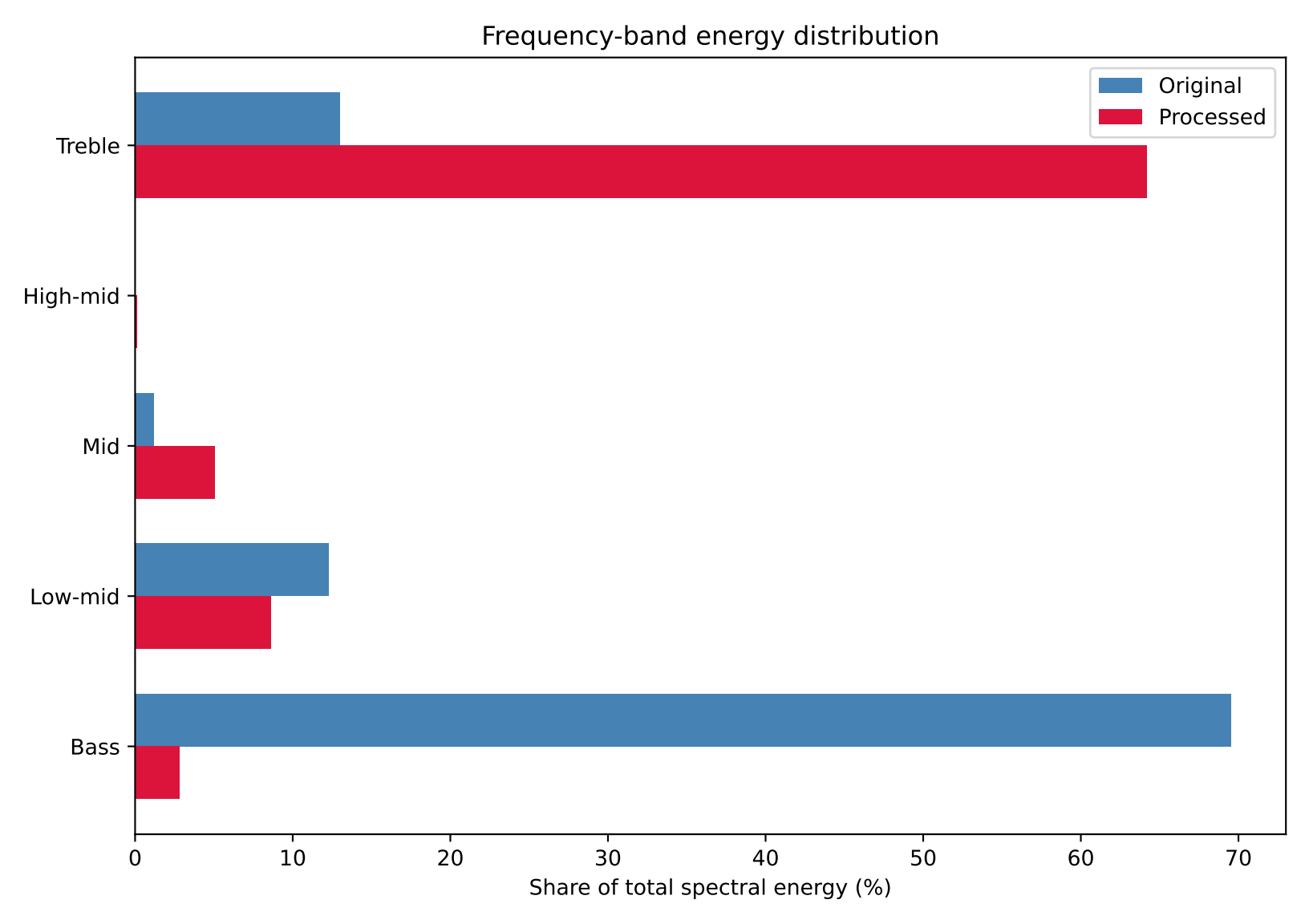
Clipping protection:
  Peak before normalization : 0.8862
  Peak after normalization  : 0.8862
  Clipping detected         : False

Quantitative metrics (Original vs Processed):

Metric                         Original     Processed
--------------------------------------------------------
Duration (s)                      20.20         20.20
RMS                               0.0600        0.0272
Peak                              0.9000        0.8862
Energy                          3210.88        658.45
Spectral Centroid (Hz)         13042.3       14328.3
Dominant Freq (Hz)               196.7         587.9
Zero-Crossing Rate               0.1039        0.1327
```


Waveform
Original
Processed
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Magnitude spectrum
Original
Processed
−40
−60
−80
−100
−120
−140
−160
10^2
10^3
10^4
Spectrogram (processed)
20000
17500
15000
12500
10000
7500
5000
2500
0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)


Parametric EQ -- Phase & Group Delay
Phase response
Phase (rad)
0.2
0.1
0.0
−0.1
−0.2
Group delay
Delay (ms)
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
0
5000
10000
15000
20000
Frequency (Hz)


Frequency-band energy distribution
Original
Processed
Treble
High-mid
Mid
Low-mid
Bass
0
10
20
30
40
50
60
70
Share of total spectral energy (%)

Filter Performance & FIR vs IIR

FIR vs IIR comparison:

Property	FIR (Graphic EQ)	IIR (Parametric EQ)
Structure	Parallel bank of 10 band-pass FIR filters	Cascade of 2 biquad section(s)
Stability	Inherently stable (no poles)	Stable (max |pole|=0.9791)
Phase	Linear (symmetric kernel)	Nonlinear (typical for IIR)
Coefficients	401	10
Processing time	512.67 ms	7.87 ms